Sugandhi K

Curriculum Vitae

Department of Computer Science
Hindustan Institute of Technology and Science (HITS), Chennai
+91-9497111459
srisugandhi.19@gmail.com
Website | Google Scholar | ORCID | Scopus

Professional Summary

Assistant Professor with research expertise in Machine Learning, Deep Learning, Computer Vision, Medical Image Processing, Human Gait Recognition and Artificial Intelligence. Recipient of the DST INSPIRE Fellowship with publications in SCI-indexed and international journals. Active reviewer for reputed journals and certified in Data Science and Artificial Intelligence through EICT Academy, IIT Guwahati.

Education

2015–2020 **Doctor of Philosophy (Ph.D.) in Computer Science**, *Department of Information Technology, Kannur University*, Kannur, Kerala, India.
INSPIRE Fellow

2013–2014 **Master of Philosophy (M.Phil.) in Computer Science**, *Department of Information Technology, Kannur University*, Kannur, Kerala, India.
CGPA: 7.93/10, Batch Topper

2011–2013 **Master of Science (M.Sc.) in Computer Science**, *Department of Computer Science, Central University of Kerala*, Kasaragod, Kerala, India.
CGPA: 8.31/10, First Rank Holder

2008–2011 **Bachelor of Science (B.Sc.) in Computer Science**, *St. Pius X College, Rajapuram, Kannur University*, Kannur, Kerala, India.
CGPA: 3.41/4

Teaching Experience

2024–Present **Assistant Professor (Selection Grade)**, *Department of Computer Science*, Hindustan Institute of Technology and Science (HITS), Chennai, Tamil Nadu.

2023–2024 **Assistant Professor**, *Department of Computer Science and Engineering*, Vignan Foundation for Science, Technology & Research (VFSTR), Guntur, Andhra Pradesh.

2022–2023 **Guest Faculty**, *Department of Computer Science*, Central University of Kerala (CUK), Kasaragod, Kerala.

Research Experience

2015–2020 **INSPIRE Fellow**, *Kannur University*.
DST-funded doctoral research in Machine Learning, Computer Vision, Human Gait Recognition, and Medical Image Processing.

Technical Skills

Programming	Python, Java, C, C++, MATLAB
Artificial Intelligence	Machine Learning, Deep Learning, Generative AI, Large Language Models (LLMs), Prompt Engineering
Machine Learning	Classification, Regression, Clustering, Ensemble Learning, Feature Engineering, Model Evaluation

Deep Learning	CNNs, Transfer Learning, Computer Vision, Image Classification, Object Detection
Data Science	Data Analytics, Data Visualization, Exploratory Data Analysis (EDA), Statistical Analysis
Tools & Frameworks	Scikit-learn, TensorFlow, Keras, Pandas, NumPy, Matplotlib, Jupyter Notebook
Research Tools	Weka, MATLAB Classification Learner, SPSS
Research Areas	Computer Vision, Medical Image Processing, Pattern Recognition, Human Gait Analysis, Sign Language Recognition
Professional Skills	Technical Writing, Scientific Communication, Collaboration, Critical Thinking, Academic Mentoring
Languages	English, Malayalam, Hindi

Publications

Journal Articles

2026 Sugandhi K., Kavya Sree Chowdary Kari. **SignFlow: A CNN Driven System for Real-Time ASL-to-Text Translation**. *International Journal of Biometrics*. DOI: 10.1504/IJBM.2026.10074006

2021 Farha Fatina Wahid, Sugandhi K., Raju G. A Fusion Based Approach for Blood Vessel Segmentation from Fundus Images by Separating Brighter Optic Disc. *Pattern Recognition and Image Analysis*, 31, 811–820. DOI: 10.1134/S105466182104026X

2021 Farha Fatina Wahid, Sugandhi K., Raju G., Debabrata Swain, Biswaranjan Acharya, Manas Ranjan Pradhan. A Novel Threshold Based Method for Vessel Intensity Detection and Extraction from Retinal Images. *International Journal of Advanced Computer Science and Applications*, 12(6), 546–554. DOI: 10.14569/IJACSA.2021.0120663

2021 Sugandhi K., Farha Fatina Wahid, Raju G. Statistical Features from Frame Aggregation and Differences for Human Gait Recognition. *Multimedia Tools and Applications*, Springer. DOI: 10.1007/s11042-021-10655-z

2020 Farha Fatina Wahid, Sugandhi K., Raju G. Cuda Implementation of Non-Local Means Algorithm for GPU Processors. *Indian Journal of Computer Science and Engineering*, 11(1), 66–75. DOI: 10.21817/indjcse/2020/v11i1/201101057

2019 Sugandhi K., Farha Fatina Wahid, Nikesh P., Raju G. An Overlap-Based Human Gait Cycle Detection. *International Journal of Biometrics*, 11(2), 148–159. DOI: 10.1504/IJBM.2019.099033

2018 Farha Fatina Wahid, Sugandhi K., Raju G. Cluster-Based Non-Local Filters for Colour Image Denoising. *International Journal of Digital Signals and Smart Systems*, 2(3), 185–201. DOI: 10.1504/IJDSSS.2018.097306

2017 Sugandhi K., Farha Fatina Wahid, Raju G. Cluster Based Non-Local Approaches for MR Image Denoising. *International Journal of Engineering and Future Technology*, 14(1), 24–38.

Book Chapters

2019 Sugandhi K., Raju G. **Discriminative Gait Features Based on Signal Properties of Silhouette Centroids**. In: Singh M., Gupta P., Tyagi V., Flusser J., Oren T., Kashyap R. (Eds.), *Advances in Computing and Data Sciences (ICACDS 2019)*, Communications in Computer and Information Science (CCIS), Vol. 1046, Springer, Singapore, pp. 680–688. DOI: 10.1007/978-981-13-9942-8_63

2018 Farha Fatina Wahid, Sugandhi K., Raju G. **Two Stage Histogram Enhancement Schemes to Improve Visual Quality of Fundus Images**. Communications in Computer and Information Science, Springer. DOI: 10.1007/978-981-13-1810-8_1

2017 Raju G., Farha Fatina Wahid, Sugandhi K. **Additive Noise Removal by Combining Non Local Means Filtering and a Local Fuzzy Filter – A Fusion Approach**. Communications in Computer and Information Science, pp. 30–39. DOI: 10.1007/978-981-10-5427-3_4

2017 Sugandhi K., Farha Fatina Wahid, Raju G. **Feature Extraction Methods for Human Gait Recognition – A Survey**. Communications in Computer and Information Science, pp. 377–385. DOI: 10.1007/978-981-10-5427-3_40

2016 Farha Fatina Wahid, Sugandhi K., Raju G. **A Review on Non Local Filtering Approaches for Medical Image Denoising**. National Conference on Recent Trends in Computational Intelligence and Image Processing (RCIP 2016).

Conference Publications

2024 Sugandhi K., R. Uday Kumar Reddy, K. Ravi Kiran Reddy and B. Bella Reddy. **Predicting Hate Words and Offensive Language: A Machine Learning Approach**. 8th International Conference on Inventive Systems and Control (ICISC), pp. 205–210. DOI: 10.1109/ICISC62624.2024.00042

2019 Sugandhi K., Farha Fatina Wahid and Raju G. **Inter Frame Statistical Feature Fusion for Human Gait Recognition**. International Conference on Data Science and Communication (IConDSC 2019). DOI: 10.1109/IconDSC.2019.8816977

2019 Sugandhi K. and Raju G. **An Efficient HOG-Centroid Descriptor for Human Gait Recognition**. Amity International Conference on Artificial Intelligence (AICAI 2019). DOI: 10.1109/AICAI.2019.8701366

2017 Farha Fatina Wahid, Sugandhi K. and Raju G. **Fusion of Local and Non-Local Filters for Denoising MRI Images**. International Conference on Infocom Technologies and Unmanned Systems (ICTUS 2017). DOI: 10.1109/ICTUS.2017.8286064

2017 Sugandhi K., Farha Fatina Wahid and Raju G. **Detection of Human Gait Cycle: An Overlap Based Approach**. International Conference on Infocom Technologies and Unmanned Systems (ICTUS 2017). DOI: 10.1109/ICTUS.2017.8286067

Patent

2025 **Granted Patent: Cooling Device**, *Patent No. 432989-001*, Controller General of Patents, Designs and Trade Marks, India.

Research Metrics

Google Scholar	Citations: 70 h-index: 6 i10-index: 1
Publications	8 Journal Articles 5 Book Chapters 5 Conference Publications
Research Profile	Active Google Scholar, ORCID, and Scopus Author Profiles

Academic Service

2024–Present	Reviewer, *Multimedia Tools and Applications (MTAP)*, Springer.
2024–Present	Reviewer, *International Journal of Image and Graphics (IJIG)*.

Awards and Honors

2017 Best Paper Award, International Conference on Infocom Technologies and Unmanned Systems (ICTUS).

2015 INSPIRE Fellowship, Department of Science and Technology (DST), Government of India.

2014 Batch Topper, M.Phil. Computer Science, Kannur University.

2013 First Rank Holder, M.Sc. Computer Science, Central University of Kerala.

2006 School Level Topper, Government High School Ambalathara, Kerala.

2006 Rajya Puraskar Award, Bharat Scouts and Guides, Kerala.

Professional Development

Conference Presentations

2019 Third International Conference on Advances in Computing and Data Sciences (ICACDS 2019), Inderprastha Engineering College, Ghaziabad, India.

2019 International Conference on Data Science and Communication (IConDSC 2019), Christ University, Bengaluru, India.

2019 Amity International Conference on Artificial Intelligence (AICAI 2019), Amity University, Dubai, UAE.

2019 National Seminar on Cyber Security and Digital Forensic (CYFO 2019), Department of Information Technology, Kannur University.

2017 International Conference on Infocom Technologies and Unmanned Systems (ICTUS 2017), Amity University, Dubai, UAE.

2016 First International Conference on Advances in Computing and Data Sciences (ICACDS 2016), Krishna Engineering College, Ghaziabad, India.

2016 National Conference on Recent Trends in Computational Intelligence and Image Processing (RCIP 2016).

Poster Presentations

2019 DST INSPIRE Fellowship Review Meeting, Gandhi Institute of Technology and Management (GITAM), Bengaluru, India.

Faculty Development Programs and Workshops

2025 NPTEL FDP Certification – Introduction to Machine Learning, IIT Madras.

2025 NPTEL FDP Certification – Data Analytics with Python, IIT Roorkee.

2025 Faculty Development Program on *Next-Gen Intelligence: AI Applications in Industry, Academia and Beyond*, St. Joseph's Institute of Technology.

2024 Faculty Development Program on *Quantum Enabled Science and Technology*, Institute of Engineering and Management (IEM), Kolkata.

2024 Faculty Development Program on *Harnessing Artificial Intelligence in High Performance Advanced Computing*, Hindustan College of Arts and Science.

2024 National Webinar on *Large Language Models for Process Automation*, CSI Kancheepuram Chapter.

2024 National Workshop on *Scopus and Web of Science Indexing Strategies for Journals, Conference Proceedings and Books*.

2024 IP Awareness Training Program, National Intellectual Property Awareness Mission, Government of India.

2024 Faculty Development Program on *Harnessing the Potential of Green AI, Machine Learning and Hardware Security*, University of Engineering and Management, Kolkata.

2023 Faculty Development Program on *AI for Building Speech and Vision Applications using Python*, Galgotias College of Engineering and Technology.

2020 National Workshop on *Role of Machine Learning and Deep Learning in Cyber Security*.

2020 International Webinar on *Next Generation Computing Paradigms*, Kannur University.

2020 International Webinar on *Recent Technologies in Computing*, Bharathiar University.

2020 International Web Conference on *Impact of COVID-19 on Education System*.

2018 Pre-conference Workshop on Image Classification, University of Mysore.

2017 National Workshop on Technologies and Tools for Scientific Research (TTSR 2017), University of Kerala.

2016 Orientation Programme on Anti-Plagiarism, Kannur University.

2014 National Seminar on Concepts and Trends in Research Methodology, Kannur University.

2013 National Conference on Pattern Analysis and Applied Intelligence (PAAI 2013), Central University of Kerala.

2013 Workshop on Discrete Mathematics – Graph Theory, Central University of Kerala.

Certifications

2025 **Advanced Certification in Data Science & Artificial Intelligence**, *E&ICT Academy, IIT Guwahati*, Besant Technologies.
Offline Program (July 2025 – December 2025)

2025 **AWS Training**, *Besant Technologies*.
Course Completion Certificate

2025 **Python Training**, *Besant Technologies*.
Course Completion Certificate

2025 **Introduction to Machine Learning**, *NPTEL, IIT Madras*.

2025 **Data Analytics with Python (Elite)**, *NPTEL, IIT Roorkee*.

2021 **100 Days of Code – Python Pro Bootcamp**, *Udemy*.

Professional References

Prof. (Dr.) Raju G.
Professor
Department of Computer Science and Engineering
Faculty of Engineering
Christ (Deemed to be University), Banglore
Phone: +91-9446106628
Email: kurupgraju@gmail.com

Prof. (Dr.) Rajesh R.
Professor
Department of Computer Science
Central University of Kerala
Phone: +91-9894850299
Email: kollamrajeshr@gmail.com

Prof. (Dr.) Jayasudha J. S.
Professor and Head
Department of Computer Science
Central University of Kerala
Phone: +91-8848991155
Email: jayasudhajs@cukerala.ac.in